Chapter 1

Example problem: Adaptive simulation of finite-Reynolds number entry flow into a 3D tube

In this example we shall demonstrate the spatially adaptive solution of the steady 3D Navier-Stokes equations using the problem of developing pipe flow.

1.1 The example problem

The 3D developing pipe flow in a quarter-tube domain.

Solve the steady Navier-Stokes equations:

$$Re\ u_j \frac{\partial u_i}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \qquad (1)$$

and

$$\frac{\partial u_i}{\partial x_i} = 0,$$

in the quarter-tube domain $D = \left\{x_1 \geq 0, x_2 \geq 0, x_1^2 + x_2^2 \leq 1, x_3 \in [0, L]\right\}$, subject to the Dirichlet boundary conditions:

$$\mathbf{u}|_{\partial D_{wall}} = (0, 0, 0), \qquad (2)$$

on the curved wall $\partial D_{wall} = \{(x_1, x_2, x_3) \mid x_1^2 + x_2^2 = 1\}$,

$$\mathbf{u} \cdot \mathbf{n}|_{\partial D_{sym[1,2]}} = 0, \qquad (3)$$

(where $\mathbf{n}$ is the outer unit normal vector) on the symmetry boundaries $\partial D_{sym[1]} = \{(x_1, x_2, x_3) \mid x_1 = 0\}$ and $\partial D_{sym[2]} = \{(x_1, x_2, x_3) \mid x_2 = 0\}$,

$$\mathbf{u}|_{\partial D} = (0, 0, 1 - (x_1^2 + x^2)^\alpha), \qquad (4)$$

on the inflow boundary, $\partial D_{inflow} = \{(x_1, x_2, x_3) \mid x_3 = 0\}$, and finally

$$u_1|_{\partial D_{outflow}} = u_2|_{\partial D_{outflow}} = 0, \qquad (5)$$

(parallel flow) on the outflow boundary $\partial D_{outflow} = \{(x_1, x_2, x_3) \mid x_3 = L\}$. Note that the axial velocity component, u_3, is not constrained at the outflow. Implicitly, we are therefore setting the axial component of traction on the fluid to zero,

$$t_3|_{\partial D_{outflow}} = \left(-p + 2\,\frac{\partial u_3}{\partial x_3}\right)\Bigg|_{\partial D_{outflow}} = 0.$$

Since $\partial u_1/\partial x_1 = \partial u_2/\partial x_2 = 0$ in the outflow cross-section [see (5], and the flow is incompressible, this is equivalent to (weakly) setting the pressure at the outflow to zero,

$$p|_{\partial D_{outflow}} = 0.$$

1.1.1 Results for Taylor-Hood elements

The figure below shows the results computed with `oomph-lib`'s 3x3x3-node 3D adaptive Taylor-Hood elements for the parameters $\alpha = 20$, $L = 7$ and $Re = 100$. The large exponent $\alpha = 20$ imposes a very blunt inflow profile, which creates a thin boundary layer near the wall. Diffusion of vorticity into the centre of the tube smooths the velocity profile which ultimately approaches a parabolic Poiseuille profile. If you are viewing these results online you will be able to see how successive mesh adaptations refine the mesh – predominantly near the entry region where the large velocity gradients in the boundary layer require a fine spatial discretisation. Note also that on the coarsest mesh, even the (imposed) inflow profile is represented very poorly.

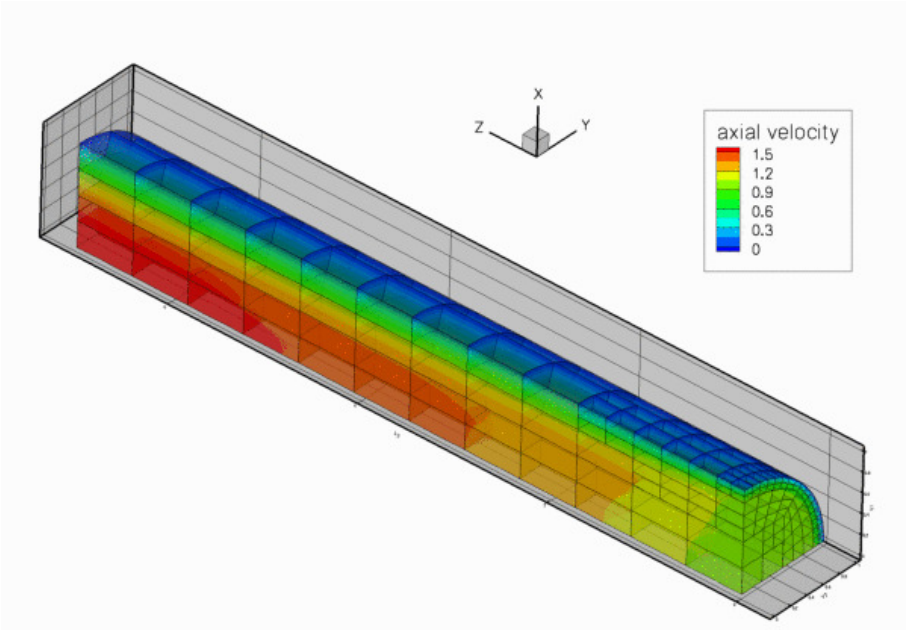

Figure 1.1 Contour plot of the axial velocity distribution for Re=100. Flow is from right to left.

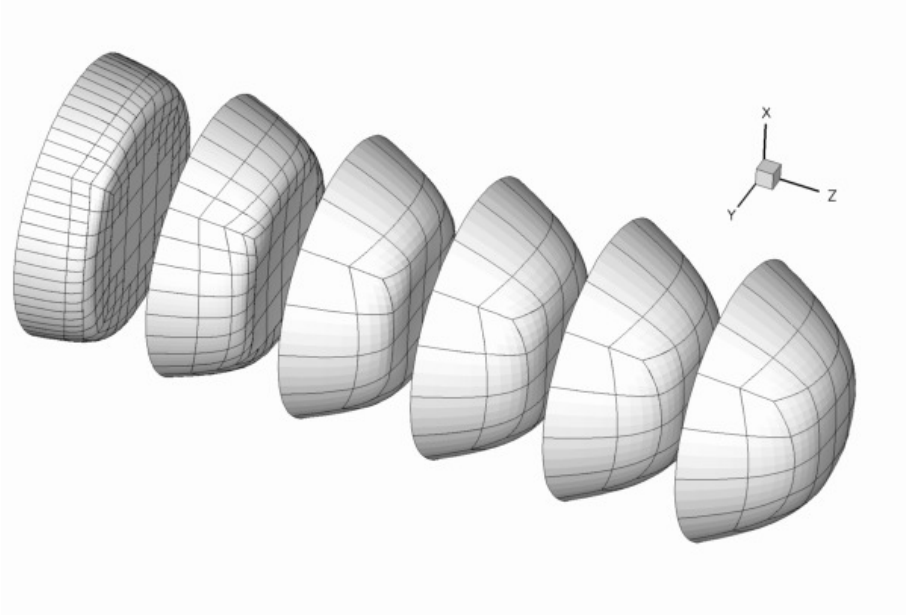

Figure 1.2 Axial velocity profiles in equally-spaced cross-sections along the tube for Re=100. Flow is from left to right.

1.2 Global parameters and functions

The problem only contains one global parameter, the Reynolds number, which we define in a namespace, as usual.

```
//=start_of_namespace==================================================
/// Namespace for physical parameters
//=====================================================================
namespace Global_Physical_Variables
{
 /// Reynolds number
 double Re=100;
} // end_of_namespace
```

1.3 The driver code

Since the 3D computations can take a long time, and since all demo codes are executed during `oomph-lib's` self-test procedures, we allow the code to operate in two modes:

- By default, we specify error targets for which the code refines the mesh near the inflow region, and allow up to five successive mesh adaptations. The code is executed in this mode if the executable is run without any command line arguments.
- If the code is run during the self-test procedure (indicated by specifying some random command line argument), we only perform one level of adaptation to speed up the self-test. However, because the original mesh is very coarse, the first mesh adaptation refines *all* elements in the mesh (cf. the animation of the adaptive mesh refinement shown above), so that no hanging nodes are created – not a good test-case for a validation run! Therefore adjust the error targets so that the first (and only) mesh adaption only refines a few elements and therefore creates a few hanging nodes.

The main code therefore starts by storing the command line arguments and setting the adaptation targets accordingly:

```
//=start_of_main==========================================================
/// Driver for 3D entry flow into a quarter tube. If there are
/// any command line arguments, we regard this as a validation run
/// and perform only a single adaptation
//========================================================================
int main(int argc, char* argv[])
{

 // Store command line arguments
 CommandLineArgs::setup(argc,argv);

 // Allow (up to) five rounds of fully automatic adapation in response to
 //-------------------------------------------------------------------------
 // error estimate
 //---------------
 unsigned max_adapt;
 double max_error_target,min_error_target;

 // Set max number of adaptations in black-box Newton solver and
 // error targets for adaptation
 if (CommandLineArgs::Argc==1)
  {
   // Up to five adaptations
   max_adapt=5;

   // Error targets for adaptive refinement
   max_error_target=0.005;
   min_error_target=0.0005;
  }
 // Validation run: Only one adaptation. Relax error targets
 // to ensure that not all elements are refined so we're getting
 // some hanging nodes.
 else
  {
   // Validation run: Just one round of adaptation
   max_adapt=1;

   // Error targets for adaptive refinement
   max_error_target=0.02;
   min_error_target=0.002;
  }
 // end max_adapt setup
```

We then create a `DocInfo` object to specify the labels for the output files, and solve the problem, first with Taylor-Hood and then with Crouzeix-Raviart elements, writing the results from the two discretisations to different directories:

```
 // Set up doc info
 DocInfo doc_info;
 // Do Taylor-Hood elements
 //------------------------
 {
  // Set output directory
  doc_info.set_directory("RESLT_TH");

  // Step number
  doc_info.number()=0;

  // Build problem
  EntryFlowProblem<RefineableQTaylorHoodElement<3> >
   problem(doc_info,min_error_target,max_error_target);
```

```
  cout « " Doing Taylor-Hood elements " « std::endl;


  // Doc solution after solving
  problem.doc_solution();

  // Solve the problem
  problem.newton_solve(max_adapt);
 }


 // Do Crouzeix-Raviart elements
 //----------------------------
 {
  // Set output directory
  doc_info.set_directory("RESLT_CR");

  // Step number
  doc_info.number()=0;

  // Build problem
  EntryFlowProblem<RefineableQCrouzeixRaviartElement<3> >
   problem(doc_info,min_error_target,max_error_target);

  cout « " Doing Crouzeix-Raviart elements " « std::endl;

  // Solve the problem
  problem.newton_solve(max_adapt);
 }

} // end_of_main
```

1.4 The problem class

The problem class is very similar to the ones used in the 2D examples. We pass the `DocInfo` object and the target errors to the `Problem` constructor.

```
//=start_of_problem_class=============================================
/// Entry flow problem in quarter tube domain
//====================================================================
template<class ELEMENT>
class EntryFlowProblem : public Problem
{

public:

 /// Constructor: Pass DocInfo object and target errors
 EntryFlowProblem(DocInfo& doc_info, const double& min_error_target,
                  const double& max_error_target);

 /// Destructor (empty)
 ~EntryFlowProblem() {}
```

The function `Problem::actions_after_newton_solve()` is used to document the solutions computed at various levels of mesh refinement:

```
 /// Doc the solution after solve
 void actions_after_newton_solve()
  {
   // Doc solution after solving
   doc_solution();

   // Increment label for output files
   Doc_info.number()++;
  }
```

The function `Problem::actions_before_newton_solve()` is discussed below, and, as in all adaptive Navier-Stokes computations, we use the function `Problem::actions_after_adapt()` to pin any redundant pressure degrees of freedom; see another tutorial for details.

```
 /// Update the problem specs before solve
 void actions_before_newton_solve();

 /// After adaptation: Pin redudant pressure dofs.
 void actions_after_adapt()
  {
   // Pin redudant pressure dofs
   RefineableNavierStokesEquations<3>::
    pin_redundant_nodal_pressures(mesh_pt()->element_pt());
  }
```

Finally, we have the usual `doc_solution()` function and include an access function to the mesh. The private member data `Alpha` determines the bluntness of the inflow profile.

```
 /// Doc the solution
```

```
 void doc_solution();

 /// Overload generic access function by one that returns
 /// a pointer to the specific  mesh
 RefineableQuarterTubeMesh<ELEMENT>* mesh_pt()
  {
   return dynamic_cast<RefineableQuarterTubeMesh<ELEMENT>*>(Problem::mesh_pt());
  }

private:

 /// Exponent for bluntness of velocity profile
 int Alpha;

 /// Doc info object
 DocInfo Doc_info;

}; // end_of_problem_class
```

1.5 The constructor

We start by building the adaptive mesh for the quarter tube domain. As for most meshes with curvilinear boundaries, the `RefineablequarterTubeMesh` expects the curved boundary to be represented by a `GeomObject`. We therefore create an `EllipticalTube` with unit half axes, i.e. a unit cylinder and a pass a pointer to the `Geom↩Object` to the mesh constructor. The "ends" of the curvilinear boundary (in terms of the maximum and minimum values of the Lagrangian coordinates that parametrise the shape of the `GeomObject`) are such that it represents a quarter of a cylindrical tube of length $L = 7$.

```
//=start_of_constructor=====================================================
/// Constructor: Pass DocInfo object and error targets
//=========================================================================
template<class ELEMENT>
EntryFlowProblem<ELEMENT>::EntryFlowProblem(DocInfo& doc_info,
                                            const double& min_error_target,
                                            const double& max_error_target)
 : Doc_info(doc_info)
{

 // Setup mesh:
 //------------

 // Create geometric objects: Elliptical tube with half axes = radius = 1.0
 double radius=1.0;
 GeomObject* Wall_pt=new EllipticalTube(radius,radius);

 // Boundaries on object
 Vector<double> xi_lo(2);
 // height of inflow
 xi_lo[0]=0.0;
 // start of Wall_pt
 xi_lo[1]=0.0;

 Vector<double> xi_hi(2);
 // height of outflow
 xi_hi[0]=7.0;
 // end of Wall_pt
 xi_hi[1]=2.0*atan(1.0);

 // # of layers
 unsigned nlayer=6;

 //Radial divider is located half-way along the circumference
 double frac_mid=0.5;

 // Build and assign mesh
 Problem::mesh_pt()=
  new RefineableQuarterTubeMesh<ELEMENT>(Wall_pt,xi_lo,frac_mid,xi_hi,nlayer);
```

Next, we build an error estimator and specify the target errors for the mesh adaptation:

```
 // Set error estimator
 Z2ErrorEstimator* error_estimator_pt=new Z2ErrorEstimator;
 mesh_pt()->spatial_error_estimator_pt()=error_estimator_pt;
 // Error targets for adaptive refinement
 mesh_pt()->max_permitted_error()=max_error_target;
 mesh_pt()->min_permitted_error()=min_error_target;
```

Now we have to apply boundary conditions on the various mesh boundaries. [**Reminder:** If the numbering of the mesh boundaries is not apparent from its documentation (as it should be!), you can use the function `Mesh↩`

`::output_boundaries(...)` to output them in a tecplot-readable form.

```
//Doc the boundaries
ofstream some_file;
char filename[100];
snprintf(filename, sizeof(filename), "boundaries.dat");
some_file.open(filename);
mesh_pt()->output_boundaries(some_file);
some_file.close();
```

If the mesh has N boundaries, the output file will contain N different zones, each containing the (x, y, z) coordinates of the nodes on the boundary.]
For the `RefineableQuarterTubeMesh`, the boundaries are numbered as follows:

- Boundary 0: "Inflow" cross section; located along the line parametrised by $x_3 = \xi_0 = \xi_0^{lo}$ on the `Geom↩Object` that specifies the wall.
- Boundary 1: Plane $x_1 = 0$
- Boundary 2: Plane $x_2 = 0$
- Boundary 3: The curved wall
- Boundary 4: "Outflow" cross section; located along the line parametrised by $x_3 = \xi_0 = \xi_0^{hi}$ on the `Geom↩Object` that specifies the wall.

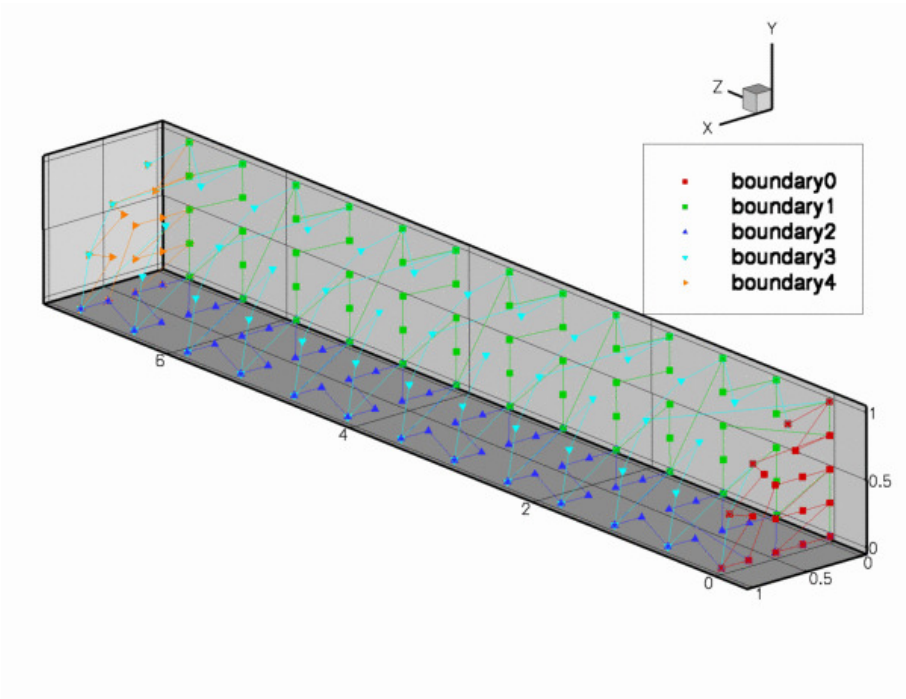

Figure 1.3 Plot of the mesh boundaries.

We apply the following boundary conditions:

- Boundary 0: ($x_3 = 0$) pin all three velocities.
- Boundary 1: ($x_1 = 0$) pin u_1.
- Boundary 2: ($x_2 = 0$) pin u_2.
- Boundary 3: ($x_1^2 + x_2^2 = 1$) pin all three velocities.
- Boundary 4: ($x_3 = L$) pin u_1 and u_2.

```
// Set the boundary conditions for this problem: All nodal values are
// free by default -- just pin the ones that have Dirichlet conditions
// here.
unsigned num_bound = mesh_pt()->nboundary();
for(unsigned ibound=0;ibound<num_bound;ibound++)
 {
  unsigned num_nod= mesh_pt()->nboundary_node(ibound);
  for (unsigned inod=0;inod<num_nod;inod++)
   {
    // Boundary 1 is the vertical symmetry boundary: We allow flow in
    //the y-direction. Elsewhere, pin the y velocity
    if(ibound!=1) mesh_pt()->boundary_node_pt(ibound,inod)->pin(1);

    // Boundary 2 is the horizontal symmetry boundary: We allow flow in
    //the x-direction. Elsewhere, pin the x velocity
    if(ibound!=2) mesh_pt()->boundary_node_pt(ibound,inod)->pin(0);

    // Boundaries 0 and 3 are the inflow and the wall respectively.
    // Pin the axial velocity because of the prescribed inflow
    // profile and the no slip on the stationary wall, respectively
    if((ibound==0) || (ibound==3))
     {
      mesh_pt()->boundary_node_pt(ibound,inod)->pin(2);
     }
   }
 } // end loop over boundaries
```

Now we assign the `re_pt()` for each element and pin the redundant nodal pressures (see `another tutorial` for details).

```
// Loop over the elements to set up element-specific
// things that cannot be handled by constructor
unsigned n_element = mesh_pt()->nelement();
for(unsigned i=0;i<n_element;i++)
 {
  // Upcast from GeneralisedElement to the present element
  ELEMENT* el_pt = dynamic_cast<ELEMENT*>(mesh_pt()->element_pt(i));

  //Set the Reynolds number, etc
  el_pt->re_pt() = &Global_Physical_Variables::Re;
 }

// Pin redudant pressure dofs
RefineableNavierStokesEquations<3>::
 pin_redundant_nodal_pressures(mesh_pt()->element_pt());
```

We provide an initial guess for the velocity field by initialising all velocity components with their Poiseuille flow values.

```
// Set Poiseuille flow as initial for solution
// Inflow will be overwritten in actions_before_newton_solve()
unsigned n_nod=mesh_pt()->nnode();
for (unsigned j=0;j<n_nod;j++)
 {
  Node* node_pt=mesh_pt()->node_pt(j);
  // Recover coordinates
  double x=node_pt->x(0);
  double y=node_pt->x(1);
  double r=sqrt(x*x+y*y );

  // Poiseuille flow
  node_pt->set_value(0,0.0);
  node_pt->set_value(1,0.0);
  node_pt->set_value(2,(1.0-r*r));
 }
```

Finally, we set the value of `Alpha`, the exponent that specifies the bluntness of the inflow profile and assign the equation numbers.

```
// Set the exponent for bluntness: Alpha=2 --> Poisseuille; anything
// larger makes the inflow blunter
Alpha=20;

//Attach the boundary conditions to the mesh
cout «"Number of equations: " « assign_eqn_numbers() « std::endl;

} // end_of_constructor
```

1.6 Actions before solve

We use the function `Problem::actions_before_newton_solve()` to re-assign the inflow boundary conditions before every solve. In the present problem this is an *essential* step because the blunt inflow profile (4) cannot be represented accurately on the initial coarse mesh (see the animation of the axial velocity profiles at the beginning of this document). As discussed in the `example that illustrates the use of`

spatial adaptivity for time-dependent problems, oomph-lib automatically **(i)** applies the correct boundary conditions for newly created nodes that are located on the mesh boundaries and **(ii)** assigns the nodal values at such nodes by interpolation from the previously computed solution. This procedure is adequate on boundaries where homogeneous boundary conditions are applied, e.g. on the curved wall, the symmetry and the outflow boundaries. However, on the inflow boundary, the interpolation from the FE representation of the blunt velocity profile (imposed on the coarse initial mesh) onto the refined mesh, does not yield a more accurate representation of the prescribed inflow profile. It is therefore necessary to re-assign the nodal values on this boundary after every adaptation, i.e. before every solve.

```
//=start_of_actions_before_newton_solve==========================================
/// Set the inflow boundary conditions
//================================================================================
template<class ELEMENT>
void EntryFlowProblem<ELEMENT>::actions_before_newton_solve()
{

 // (Re-)assign velocity profile at inflow values
 //---------------------------------------------

 // Setup bluntish parallel inflow on boundary 0:
 unsigned ibound=0;
 unsigned num_nod= mesh_pt()->nboundary_node(ibound);
 for (unsigned inod=0;inod<num_nod;inod++)
  {
   // Recover coordinates
   double x=mesh_pt()->boundary_node_pt(ibound,inod)->x(0);
   double y=mesh_pt()->boundary_node_pt(ibound,inod)->x(1);
   double r=sqrt(x*x+y*y);

   // Bluntish profile for axial velocity (component 2)
   mesh_pt()->boundary_node_pt(ibound,inod)->
    set_value(2,(1.0-pow(r,Alpha)));
  }
```

1.7 Post processing

This function remains exactly the same as in the 2D examples.

```
//=start_of_doc_solution=========================================================
/// Doc the solution
//================================================================================
template<class ELEMENT>
void EntryFlowProblem<ELEMENT>::doc_solution()
{
 ofstream some_file;
 char filename[100];

 // Number of plot points
 unsigned npts;
 npts=5;

 // Output solution
 snprintf(filename, sizeof(filename), "%s/soln%i.dat",Doc_info.directory().c_str(),
         Doc_info.number());
 some_file.open(filename);
 mesh_pt()->output(some_file,npts);
 some_file.close();

} // end_of_doc_solution
```

1.8 Comments and exercises

1. Suppress the reassignment of the prescribed inflow profile in `Problem::actions_before_newton↩_solve()` to confirm that this step is essential if the computation is to converge to the exact solution.

2. Suppress the specification of the parabolic (Poiseuille) velocity profile as the initial guess for the velocity field in the problem constructor to confirm that the assignment of a "good" initial guess for the solution is essential for the convergence of the Newton method. [Hint: You can simply comment out the initialisation of the velocities – they then retain their default initial values of 0.0. When you re-run the code, the Newton iteration will "die" immediately with an error message stating that the maximum residual exceeds the default threshold of 10.0, stored in the protected data member `Problem::Max_residuals`. Try increasing the value of this threshold in the Problem constructor. Is this sufficient to make the Newton method converge?]

1.9 Source files for this tutorial

- The source files for this tutorial are located in the directory:

  demo_drivers/navier_stokes/three_d_entry_flow/

- The driver code is:

  demo_drivers/navier_stokes/three_d_entry_flow/three_d_entry_flow.cc

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1.10 PDF file

A pdf version of this document is available. \